

AI-POWERED NLP-BASED SENTIMENT ANALYSIS SYSTEM

Software Requirements Specification (SRS)

Document Type	Software Requirements Specification
Project Domain	Artificial Intelligence / Natural Language Processing
Primary Application	Text Sentiment Classification
Analysis Approach	Machine Learning (Naive Bayes) and Deep Learning (Transformers)
Document Version	1.0
Status	Requirements Baseline

Purpose: This document defines the functional, non-functional, technical, security, data, machine-learning, evaluation, and deployment requirements for the AI-Powered NLP-Based Sentiment Analysis System using Naive Bayes and Transformers.

1. Introduction

Sentiment analysis is a major natural language processing task in which user-generated text such as reviews, comments and posts is examined to determine the emotional tone expressed by its author, typically classified as positive or negative. Traditional dictionary-based, rule-based and manual annotation approaches are limited against informal, colloquial, slang-heavy or sarcastic text that does not appear in standard lexicons. Machine-learning and deep-learning techniques can learn patterns from labeled positive and negative text and automate sentiment classification, while transformer-based models can capture contextual meaning beyond individual words.

The proposed system focuses on automated sentiment classification using classical machine-learning (Naive Bayes) and transformer-based deep-learning techniques. Research in IEEE Access has demonstrated that hybrid dictionary-machine-learning systems achieve strong performance on short informal text, while comparative benchmarking studies have shown that transformer models and large language models differ substantially in in-domain versus cross-domain settings. Recent research has also emphasized topic-model-driven sentiment analysis of social media discourse, early generalization analysis, and explainability of model decisions.

The system is intended to provide a practical sentiment analysis platform while also supporting reproducible experimentation and comparative evaluation of classical and transformer-based machine-learning approaches.

2. Purpose

The purpose of the system is to determine whether a submitted text is likely to express a positive or negative sentiment using machine-learning-based analysis.

The system shall transform a text into a structured representation of features, apply a trained classification model (Naive Bayes or transformer-based), and return a prediction with an associated confidence score.

The system shall also provide an experimental framework for comparing machine-learning algorithms and evaluating their ability to generalize to previously unseen text domains.

3. System Scope

3.1 In Scope

- Text submission and validation
- Text preprocessing and normalization
- Lexical text feature extraction
- Dictionary/lexicon-based feature extraction
- N-gram and statistical feature extraction
- Contextual embedding extraction (transformer pathway)
- Feature preprocessing and selection
- Machine-learning model training (Naive Bayes)
- Deep-learning model training (transformer fine-tuning)
- Machine-learning model comparison

- Sentiment probability estimation
- Sentiment category classification
- Prediction explanation
- Model evaluation
- Cross-dataset validation
- Web-based analysis interface
- REST API
- Prediction logging
- Performance monitoring

3.2 Out of Scope

- Guaranteed detection of every sentiment expression including sarcasm and irony
- Real-time streaming analysis of live social media feeds in the initial release
- Automatic posting or modification of content on third-party platforms
- Multi-label and aspect-based sentiment analysis in the initial release
- Multilingual sentiment analysis beyond the selected language(s) in the initial release
- Human-level understanding of implicit, context-dependent or culturally specific sentiment
- Automatic annotation of unlabeled text at scale

4. System Overview

The system follows a layered pipeline architecture:

User → Web Interface → API Layer → Text Validation → Text Preprocessing → Feature Extraction → Feature Selection/Transformation → Machine Learning Model → Prediction Probability → Sentiment Assessment → Result/Report

The architecture separates text processing, feature engineering, machine learning, prediction, and presentation so that individual components can be modified or evaluated independently. The Naive Bayes pathway operates on lexical, dictionary and n-gram features, while the transformer pathway operates on contextual token representations produced by a pre-trained language model.

5. Functional Requirements

FR-001 — Text Submission

The system shall provide an interface through which a user or client application can submit a text for analysis.

FR-002 — Text Validation

The system shall validate submitted text before analysis and shall reject empty, malformed, unsupported, or invalid input.

FR-003 — Text Normalization

The system shall normalize text into a consistent representation (lowercasing, punctuation handling, tokenization, stop-word policy) before feature extraction.

FR-004 — Feature Extraction

The system shall extract measurable characteristics from the text, including lexical, dictionary-based, n-gram, and contextual embedding characteristics where applicable.

FR-005 — Feature Processing

The system shall preprocess extracted features using the same transformation pipeline used during model training.

FR-006 — Feature Selection

The system shall support feature selection or dimensionality reduction when required by the selected experimental configuration.

FR-007 — Model Loading

The production inference service shall load a validated and versioned machine-learning model.

FR-008 — Sentiment Classification

The system shall classify a submitted text as positive or negative. A configurable intermediate neutral/mixed category may be supported.

FR-009 — Probability Estimation

The system shall return a probability or confidence score associated with the sentiment prediction.

FR-010 — Sentiment Category Classification

The system shall map the model output to a configurable sentiment category (e.g., strongly negative, negative, neutral, positive, strongly positive).

FR-011 — Prediction Explanation

The system shall expose important features or explanatory information associated with the prediction where supported by the selected model.

FR-012 — Prediction Record

The system shall optionally record the analysis identifier, timestamp, model version, prediction, and relevant evaluation metadata.

FR-013 — Report Generation

The system shall support generation of an analysis report containing the text analysis result and associated model metadata.

FR-014 — Error Handling

The system shall return meaningful error responses for invalid input, missing features, unavailable models, and internal processing failures.

6. Text Feature Requirements

The feature extraction component shall support a configurable feature set. The final feature set shall be documented and versioned with the model.

Category	Example Features
Text lexical	Text length, word count, sentence count, average word length, unique word ratio
Character	Exclamation marks, question marks, capitalization ratio, special characters, emoticons
Dictionary/Lexicon	Positive word count, negative word count, negation presence, degree adverb counts
Structure	Sentence count, punctuation density, token patterns, hashtag/mention counts
N-gram	Unigram, bigram and trigram token patterns
Statistical	TF-IDF weights, term frequencies, entropy and character-distribution features
Contextual	Token-level contextual embeddings from a pre-trained transformer model

Feature definitions shall be documented so that the training and inference pipelines produce identical feature vectors for both the Naive Bayes (feature-based) and transformer (embedding-based) pathways.

7. Machine Learning Requirements

ML-001 — Training Pipeline

The system shall provide a reproducible training pipeline covering data loading, preprocessing, feature preparation, model training, and evaluation.

ML-002 — Baseline Models

The experimental framework shall support baseline models such as Naive Bayes (Multinomial/Bernoulli with Laplace smoothing), Logistic Regression, Decision Tree, Random Forest, SVM, or equivalent baseline classifiers.

ML-003 — Transformer Models

The framework shall support transformer-based models, including fine-tuned BERT, DistilBERT, RoBERTa or equivalent pre-trained language models, for comparative evaluation.

ML-004 — Hyperparameter Optimization

The framework shall support configurable hyperparameter optimization using an appropriate search strategy for both classical and transformer models.

ML-005 — Model Versioning

Every production model shall have an identifiable version associated with its training configuration and feature schema.

ML-006 — Reproducibility

Training experiments shall record dataset versions, feature versions, model parameters, evaluation configuration, and random seeds where applicable.

8. Dataset Requirements

DATA-001 — Dataset Documentation

Every dataset shall have documented source, collection information, class definitions, feature definitions, licensing or usage conditions, and preprocessing steps.

DATA-002 — Class Labels

The dataset shall distinguish positive and negative sentiment samples using well-defined labels; a neutral class may be defined where applicable.

DATA-003 — Data Quality

The preprocessing pipeline shall identify duplicate records, missing values, malformed text, inconsistent labels, and other relevant data-quality issues.

DATA-004 — Class Distribution

The class distribution shall be measured and reported. Appropriate balancing or weighting techniques shall be applied where necessary.

DATA-005 — Data Leakage Prevention

Training, validation, and test data shall be separated in a manner that minimizes duplicate or near-duplicate samples leaking between splits.

9. Cross-Dataset Validation Requirements

Cross-dataset validation is a core research requirement. The purpose is to determine whether the sentiment analysis model generalizes beyond the dataset used for model training.

CV-001 — Independent Dataset Testing

The system shall support evaluation on at least one dataset that is independent of the primary training dataset.

CV-002 — Dataset-Specific Results

Performance results shall be reported separately for each evaluation dataset.

CV-003 — Generalization Analysis

The system shall compare within-dataset and cross-dataset performance and identify significant performance degradation.

CV-004 — Feature Stability

The analysis shall investigate whether the selected features remain effective across different text domains and platforms.

This requirement is motivated by recent research showing that strong in-domain performance does not necessarily translate to equivalent performance on independent datasets or domains. The 2024 IEEE

Access comparative study demonstrated that fine-tuned transformer models excel in their training domains while zero-shot large language models can fail sharply on implicit, multi-aspect data, and the 2020 IEEE Access study achieved 88.2% accuracy on a single-platform danmaku corpus without independent cross-dataset validation. Trained-model results remain tied to the benchmark datasets used, and cross-domain degradation must be explicitly quantified.

10. Model Evaluation Requirements

Metric	Purpose
Accuracy	Overall classification correctness
Precision	Reliability of positive/negative predictions
Recall	Ability to identify samples of each sentiment class
F1-score	Balance between precision and recall
Specificity	Ability to correctly identify samples of the opposite class
False Positive Rate	Rate of negative texts incorrectly classified as positive
False Negative Rate	Rate of positive texts missed by the model
ROC-AUC	Ranking/discrimination performance
Inference Time	Operational prediction latency (per model family)

EVAL-001 — Confusion Matrix

The system shall generate confusion matrices for applicable classification tasks.

EVAL-002 — Model Comparison

The experimental framework shall compare models using consistent datasets, features, partitions, and evaluation metrics.

EVAL-003 — Error Analysis

The evaluation shall investigate representative false-positive and false-negative cases at the linguistic level (e.g., sarcasm, negation, mixed sentiment).

EVAL-004 — Benchmarking

The final model selection shall consider detection performance, generalization, computational requirements, and deployment constraints.

11. Explainability Requirements

XAI-001 — Feature Importance

The system shall expose feature-importance information when supported by the selected model (e.g., per-class log-probability ratios for Naive Bayes, or attention/gradient-based attributions for transformers).

XAI-002 — Prediction Explanation

The system should provide human-readable information describing important characteristics associated with a prediction, such as the influential words or phrases identified by dictionary-based or feature-importance methods.

XAI-003 — Explanation Consistency

Explanations shall correspond to the model version and feature schema used to produce the prediction.

12. Web Application Requirements

WEB-001 — Analysis Interface

The application shall provide a web interface for submitting a text for sentiment analysis.

WEB-002 — Result Display

The interface shall display prediction, probability/confidence score, sentiment category, and relevant explanatory information.

WEB-003 — API Integration

The web interface shall communicate with the backend through a defined REST API.

WEB-004 — Responsive Interface

The interface shall remain usable on common desktop and mobile screen sizes.

WEB-005 — Failure Feedback

The interface shall provide understandable feedback when analysis fails.

13. API Requirements

Endpoint	Method	Purpose
/api/health	GET	Service health check
/api/analyze	POST	Analyze submitted text
/api/models	GET	Return available model metadata
/api/reports/{id}	GET	Retrieve analysis report

API-001 — Input Validation

All API inputs shall be validated before processing.

API-002 — Structured Responses

The API shall return consistent machine-readable responses.

API-003 — Error Codes

The API shall provide appropriate HTTP status codes and meaningful error messages.

14. Non-Functional Requirements

14.1 Performance

NFR-001 — Inference Performance

The production inference pipeline should provide interactive response times under normal operating conditions. The actual measured latency shall be reported as an experimental benchmark for both the Naive Bayes and transformer pathways.

14.2 Reliability

NFR-002 — Fault Tolerance

Invalid text, malformed input, missing features, model failures, and temporary service failures shall be handled without crashing the system.

14.3 Maintainability

NFR-003 — Modular Architecture

Frontend, API, feature extraction, model inference, data access, and evaluation components shall be modular.

14.4 Scalability

NFR-004 — Extensibility

The architecture shall permit additional datasets, feature sets, models, and analysis methods to be added without redesigning the entire system.

14.5 Portability

NFR-005 — Deployment Portability

The application should support deployment on standard Linux, macOS, or Windows development environments and suitable cloud platforms.

15. Security Requirements

SEC-001 — Input Sanitization

User-provided data shall be validated and safely handled before being processed by application components.

SEC-002 — No Automatic External Action

The initial text-analysis pipeline shall not automatically post, send, or otherwise act on behalf of the user on any external platform based on a submitted text.

SEC-003 — API Protection

Production API endpoints shall implement appropriate authentication, rate limiting, input validation, and access controls where applicable.

SEC-004 — Sensitive Configuration

Secrets, API keys, database credentials, and signing keys shall not be hard-coded in source code.

SEC-005 — Logging

Security-relevant application errors and operational events shall be logged without unnecessarily storing sensitive user content.

16. Privacy Requirements

PRIV-001 — Data Minimization

Only information required for analysis, evaluation, auditing, or explicitly enabled reporting functionality shall be stored.

PRIV-002 — Retention

Stored texts and analysis records shall have a configurable retention period.

PRIV-003 — Access Control

Stored analysis data shall only be accessible to authorized components and users.

17. Hardware Requirements

Component	Minimum	Recommended
CPU	4 cores	6–8 cores
RAM	8 GB	16 GB+
Storage	20 GB	50 GB+
GPU	Not required for Naive Bayes inference	Recommended for transformer fine-tuning
Network	Internet connection	Stable broadband

GPU acceleration is not a mandatory requirement for the Naive Bayes and classical feature-based pipeline; CPU-based training and inference are expected to be sufficient. GPU acceleration is recommended but not mandatory for transformer fine-tuning; CPU-based inference of a distilled transformer is acceptable for low-throughput deployments.

18. Software Requirements

Category	Recommended Technology
Language	Python 3.x
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
NLP Preprocessing	NLTK, spaCy, Regex
Transformers	Hugging Face Transformers, PyTorch or TensorFlow

Visualization	Matplotlib, Seaborn
Backend	FastAPI
Frontend	HTML, CSS, JavaScript
Database	MongoDB or PostgreSQL
Model Serialization	Joblib (classical), Safetensors/PyTorch (transformers)
Version Control	Git

19. System Constraints

- Analysis quality depends on the quality and representativeness of training data.
- The model cannot guarantee detection of every sentiment expression, particularly sarcasm, irony and implicit opinion.
- Dataset imbalance may influence evaluation results.
- Cross-dataset performance may be lower than within-dataset performance due to domain and platform shift.
- Text-only analysis cannot capture audio, visual or behavioral signals that carry sentiment.
- Transformer models require substantially more compute and labeled data than classical models, trading label cost for contextual accuracy.
- External pre-trained model repositories may impose download size and availability constraints.
- Model predictions should be treated as automated sentiment assessments rather than absolute judgments of author intent.

20. Future Extensions

- Neutral and multi-class (e.g., 7-dimension) sentiment analysis
- Aspect-based sentiment analysis (ABSA)
- Sarcasm and irony detection
- Multi-lingual and code-mixed sentiment analysis
- Zero-shot large language model comparison
- Topic-model-driven sentiment analysis (LDA, NMF, BERTopic)
- Dynamic time-distribution sentiment visualization
- Browser extension and social media integration
- Continuous model retraining and online learning
- Adversarial robustness testing against perturbed text

21. Acceptance Criteria

- A valid text can be submitted through the application.
- Invalid input is rejected safely.

- The text is converted into the required feature representation for both model families.
- A trained model produces a classification.
- A sentiment probability or confidence score is returned.
- A configurable sentiment category is displayed.
- The model version is identifiable.
- Model performance is measured using multiple appropriate metrics.
- At least one independent dataset is available for cross-dataset validation.
- False-positive and false-negative cases are analyzed.
- The API returns structured responses.
- The web interface displays analysis results correctly.
- The system handles application errors without terminating unexpectedly.
- The final model and feature pipeline are reproducible from documented artifacts.

22. Requirements Traceability Matrix

Requirement Area	Implementation Component	Validation
Text Input	Frontend + API	Input test cases
Text Validation	Validation module	Valid/invalid text tests
Feature Extraction	Feature pipeline	Feature unit tests
Model Training	Training pipeline	Training experiment
Model Comparison	Evaluation pipeline	Benchmark results
Prediction	Inference service	Prediction tests
Explainability	XAI module	Feature importance
Cross-Dataset	Evaluation pipeline	Independent test dataset
Security	API/security layer	Security test cases
Performance	Inference service	Latency benchmark
Reporting	Report module	Generated report validation

23. Research References

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